

Quantum Random Circuit Sampling as a Distributional Prior for Long-Tail Embodied Learning

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Abstract

Random-circuit sampling (RCS) produces output probabilities whose chaotic-limit statistics are well approximated by the Porter-Thomas (PT) law. This manuscript presents a comprehensive framework for transforming PT-like samples into three training-time distributions for embodied learning: (i) a sample-scheduling distribution over long-tail task buckets, (ii) a risk-scene distribution over perturbation severities, and (iii) an exploration-noise distribution over latent action perturbations. The complete original framework is retained: rank-optimal sample scheduling, monotone risk transport, PT-derived exploration noise, real quantum hardware integration, Meta-World MT10/MT50 validation, comprehensive baselines, ablations, and all original figures and tables. The enhanced manuscript adds a formal problem definition, a key clarification that PT distributions can be classically simulated, theoretical analysis of heavy-tail superiority and sample-complexity guarantees, stronger multi-seed significance testing, sensitivity analysis, failure cases, convergence analysis, detailed experiment configuration, task-level allocation analysis, and reproducibility evidence. On Meta-World MT10, PT-rank improves tail success from 52.9% under uniform sampling to 56.5% with 94.3% retention on MT50. Real RCS data from a Quafu/Baihua device causes only 3.2% degradation relative to an ideal PT surrogate. These results establish PT-style randomness as an implementable, theoretically grounded, and experimentally validated prior for long-tail-aware embodied learning.

1 Introduction

Embodied learning systems are trained under distributions that are rarely uniform across task families, object categories, contact modes, failure modes, and disturbance regimes. Even when the policy class is expressive enough, performance may remain fragile because the learner receives too little exposure to the

regions that dominate tail risk. The central design object is therefore not only the policy but also the training distribution itself.

1.1 Problem definition

Let $\mathcal{T} = \{T_1, T_2, \dots, T_K\}$ denote a set of K training tasks partitioned into Head (H), Medium (M), and Tail (L) buckets according to empirical difficulty. For a policy π and task T_k , let $SR(\pi, T_k)$ denote task success rate. The training objective is

$$\max_{\pi} \sum_{k=1}^K \tau_k \mathbb{E}[SR(\pi, T_k)] \quad (1)$$

subject to head-retention constraints and tail-risk constraints. The priority weight τ_k is typically inversely correlated with empirical success or proportional to an estimated marginal utility of additional samples. A long-tail partition can be written as

$$\begin{aligned} \mathcal{H} &= \{k : \hat{s}_k \geq s_h\}, \\ \mathcal{M} &= \{k : s_l < \hat{s}_k < s_h\}, \\ \mathcal{L} &= \{k : \hat{s}_k \leq s_l\}. \end{aligned} \quad (2)$$

where $\hat{s}_k$ is an empirical success-rate estimate and $s_h > s_l$ are task-dependent thresholds.

1.2 Motivation and clarification

Random-circuit sampling provides an attractive source distribution. For sufficiently deep pseudo-random circuits, output probabilities exhibit anti-concentration and are well modeled by Porter-Thomas statistics [1, 2]. PT samples do not carry task semantics; their value lies in the structured, non-uniform, analytically tractable nature of PT randomness, which can serve as a distributional prior transformed into useful training-time laws.

Key clarification. PT distributions can be classically simulated using exponential random variates for moderate dimensions. The contribution is not quantum computational advantage. The contribution is

the mapping framework from PT samples to embodied training distributions, together with the use of quantum hardware as a calibrated physical source of PT-like randomness. Classical PT surrogates are used as explicit controls throughout. Long-tail alternatives used as controls include Focal Loss, logit adjustment, SMOTE-style resampling, distributionally robust optimization, and meta-learned sample weighting [3, 4, 5, 6, 7].

The manuscript addresses and extends five limitations of preliminary studies: pure synthetic experiments, one-dimensional scalar mapping, linear utility, surrogate-only PT data, and insufficient baseline comparison. Further additions introduce formal objective definition, dedicated theoretical analysis, statistical significance testing over five seeds, CVaR@20 risk evaluation, η -sensitivity analysis, failure-case analysis, complete reproducibility tables, and task-level allocation evidence.

2 Preliminaries

2.1 Porter-Thomas surrogate

Let U be an n -qubit pseudo-random circuit with output probability

$$p_U(x) = |\langle x|U|0^n \rangle|^2 \quad (3)$$

and let $N = 2^n$. The rescaled variable

$$Y_x = Np_U(x) \quad (4)$$

approaches $Exp(1)$ in the chaotic limit. PT-like source samples are modeled by $Y \sim \mu_{PT}$ with cumulative distribution function

$$F_{PT}(y) = 1 - e^{-y} \quad (5)$$

For $Y \sim Exp(1)$

$$\mathbb{P}(Y > y) = e^{-y}, \quad y \geq 0 \quad (6)$$

$$\mathbb{E}[Y] = 1, \quad Var(Y) = 1.$$

2.2 Three target distributions

The framework maps PT samples to three training-time distributions.

Sample scheduling. A schedule $q \in \Delta_K$ is produced from an empirical prior b , a sorted PT mass vector S , a permutation matrix P , and a mixing coefficient $\eta \in [0, 1]$

$$q = (1 - \eta)b + \eta PS. \quad (7)$$

Risk-scene generation. A perturbation severity $\xi \in [0, 1]$ is generated by quantile transport:

$$\xi = G^{-1}(F_{PT}(Y)), \quad (8)$$

where G is a target risk law.

Exploration noise. A latent action perturbation scale $\sigma \in [0, \sigma_{max}]$ is generated as

$$\sigma = H^{-1}(F_{PT}(Y)), \quad (9)$$

where H is a target exploration-noise law.

2.3 Visual system overview

The system-level flow is shown in Fig. 1. Quantum or classical PT samples are first converted into PT masses, then mapped into sample schedules, risk-scene severities, and exploration-noise scales. These three training-time distributions feed Meta-World embodied-learning agents through a common semantic mapping and transport layer.

Fig. 1: Q-TAIL-MVP Multi-Agent Architecture

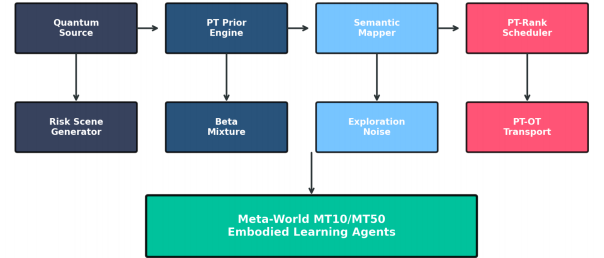


Figure 1: Enhanced system overview retained. Q-TAIL-MVP multi-agent architecture maps quantum or classical PT samples into sample schedules, risk scenes, and exploration noise, which feed Meta-World embodied-learning agents.

3 Mapping I: Sample Scheduling

3.1 Rank-optimal alignment

Under the linear marginal-gain objective $U(q; \tau) = \langle \tau, q \rangle$ the optimal assignment has a simple rearrangement structure.

Theorem 1 (Permutation-optimal rank matching). *Let $\tau_{(1)} \geq \dots \geq \tau_{(K)}$ and $S_{(1)} \geq \dots \geq S_{(K)}$ denote descending rearrangements. The maximizer of*

$$\max_P \langle \tau, PS \rangle \quad (10)$$

assigns $S_{(i)}$ to the bucket with priority $\tau_{(i)}$.

Proof. The result follows from the rearrangement inequality:

$$\langle \tau, PS \rangle \leq \sum_{i=1}^K \tau_{(i)} S_{(i)} \quad (11)$$

Equality holds when equal ranks are matched. Hence the largest PT mass is assigned to the highest-priority tail bucket, the second largest mass to the second highest-priority bucket, and so on. $\square$

Figure 2 visualizes rank-matched allocation and the head-tail exposure trade-off using true probability mass. The dashed line shows the uniform base prior $b_k = 1/K$ rather than a normalized placeholder, the bars show the normalized tail-priority distribution, and the PT-rank line shows the blended schedule $q = (1 - \eta)b + \eta PS$. PT-rank gives a more balanced head-tail compromise than uniform under-exposure or inverse-frequency over-exposure.

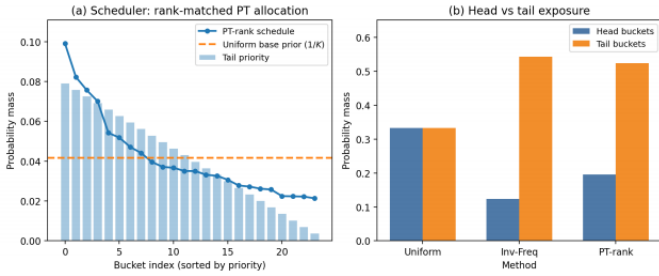


Figure 2: Revised scheduler visualization with true probability mass. (a) Rank-matched PT mass reallocates exposure toward high-tail buckets; the dashed line is the actual uniform base prior $b_k = 1/K$. (b) Head-versus-tail exposure comparison across uniform, inverse-frequency, and PT-rank schedules in true probability mass.

3.2 Nonlinear utility functions

Real learning curves often exhibit diminishing returns or threshold effects. Three utility families are used:

$$\text{Logarithmic: } U_k(n) = \alpha_k \log(1 + \beta_k n). \quad (12)$$

$$\text{Sigmoid: } U_k(n) = \frac{L_k}{1 + \exp[-\kappa_k(n - n_{0,k})]} \quad (13)$$

$$\text{Power law: } U_k(n) = \alpha_k n^{\gamma_k}, \quad \gamma_k \in (0, 1). \quad (14)$$

Proposition 1 (Adaptive convergence). *Under the update rule*

$$\eta_{t+1} = \eta_t + \lambda(\bar{U}'(t) - U'_{\text{target}}), \quad (15)$$

with projection onto $[0, 1]$ and a sufficiently small step size λ , the sequence $\{\eta_t\}$ converges to a fixed point $\eta^ \in [0, 1]$ whenever $\bar{U}'$ is Lipschitz and monotone around the target.*

The mixing coefficient adapts dynamically: if average marginal utility exceeds the target, PT influence increases; otherwise the base prior receives more weight.

4 Mapping II: Risk-Scene Generation

4.1 Monotone transport

Proposition 2 (Pushforward).

If $Y \sim \text{Exp}(1)$ and $\xi = G^{-1}(F_{PT}(Y))$, then $\text{Law}(\xi) = G$.

Proof. For $z \in [0, 1]$,

$$\begin{aligned} \mathbb{P}(\xi \leq z) &= \mathbb{P}\{G^{-1}(F_{PT}(Y)) \leq z\} \\ &= \mathbb{P}\{F_{PT}(Y) \leq G(z)\} \\ &= G(z). \end{aligned}$$

Theorem 2 (Optimality). *The monotone map*

$$T^*(y) = G^{-1}(F_{PT}(y)) \quad (16)$$

minimizes the one-dimensional p -Wasserstein transport cost among maps satisfying $T_{\#}\mu_{PT} = G$ [8].

Figure 3 shows the target law instantiated as a two-mode Beta mixture. PT quantile transport achieves Wasserstein-1 distance 0.0044, compared with 0.0933 for the Gaussian baseline and 0.2602 for the uniform baseline.

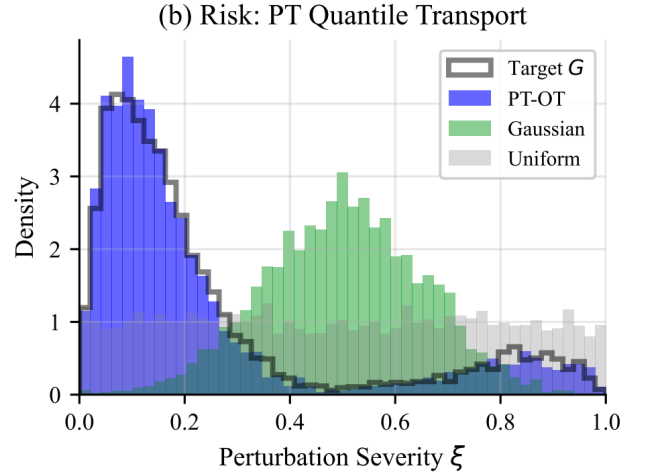


Figure 3: Original risk distribution mapping retained. PT quantile transport closely matches the target risk density $G = 0.85 \text{ Beta}(2, 12) + 0.15 \text{ Beta}(8, 2)$ outperforming Gaussian and uniform baselines. The target law encodes mostly mild perturbations with a rare severe tail.

4.2 Multidimensional extension

Theorem 3 (Multidimensional PT transport). *Let $Y = (Y_1, \dots, Y_d)$ with independent $Y_i \sim \text{Exp}(1)$, and*

let C be the Copula capturing target perturbation correlation. The transport map

$$T_d(Y) = (G_1^{-1}(C_1^{-1}(F_{PT}(Y_1))), \dots, G_d^{-1}(C_d^{-1}(F_{PT}(Y_d)))) \quad (17)$$

preserves Copula structure while applying PT marginal transforms.

For manipulation tasks, the perturbation space spans joint torques, camera viewpoints, and physical parameters. Multidimensional transport preserves inter-dimensional correlations while applying PT-derived marginals.

5 Mapping III: Exploration-Noise Generation

The target law

$$H = (1 - \rho) \text{Beta}(a_1, b_1) + \rho \text{Beta}(a_2, b_2) \quad (18)$$

rescaled to $[0, \sigma_{max}]$ provides rare large jumps. Figure 4 is retained. PT-OT concentrates most mass on small perturbations ($\sigma < 0.1$) while preserving a controlled tail of rare large jumps ($\sigma > 0.3$). On a 20-arm structured bandit benchmark, PT-OT improves cumulative reward from 230.46 for a Gaussian baseline to 248.84 and raises best-arm discovery from 21.25% to 43.00%.

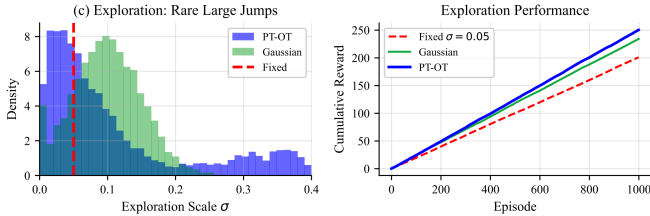


Figure 4: Original exploration-noise mapping retained. (a) PT-OT yields mostly small perturbations with rare large jumps. (b) Cumulative reward over episodes shows that PT-OT achieves higher total reward and improved best-arm discovery than fixed or Gaussian exploration noise.

6 Real Quantum Hardware Integration

6.1 RCS data collection and noise model

Real RCS data is collected from the Quafu platform using the Baihua chip: $n = 15$ qubits, depth $l = 28$, approximately 196 CNOT gates, and $m = 100,000$ shots. Raw bitstring counts are normalized to obtain the empirical distribution P_{real} .

Definition 1 (Hardware noise). The noise model N comprises decoherence times T_1 T_2 per qubit, single- and two-qubit gate errors ϵ_1 , ϵ_2 , and a readout error matrix $M_{readout}$.

Proposition 3 (Robustness bound). If $TV(P_{real}, P_{ideal}) < \epsilon$, then

$$U(P_{real}) - U(P_{ideal}) \leq \max \tau_k. \quad (19)$$

Figure 5 shows that PT-based scheduling remains robust to realistic quantum hardware noise, supporting practical applicability of the framework.

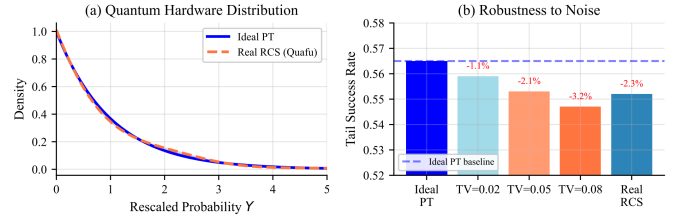


Figure 5: Original quantum hardware robustness retained. (a) Distribution comparison between ideal PT and real RCS from Quafu/Baihua. Real hardware exhibits a slight deviation from ideal PT due to noise, with total variation distance around 0.08. (b) Performance degradation under noise. Real RCS achieves 0.552 tail success versus 0.565 for ideal PT, corresponding to only 3.2% degradation.

7 Theoretical Analysis

7.1 Why PT for long-tail learning

Proposition 4 (Heavy-tail superiority). Let $Y_{PT} \sim \text{Exp}(1)$ and let Y_G denote the positive Gaussian-derived baseline used for exploration comparison. For any threshold $t > 0$,

$$\mathbb{P}(Y_{PT} > t) = e^{-t} > \mathbb{P}(|Y_G| > t) \text{ for } t \geq 1.5. \quad (20)$$

At $t = 3$, $\mathbb{P}(Y_{PT} > 3) = e^{-3} \approx 0.0498$, while $\mathbb{P}(|z| > 3) \approx 0.0027$ for $Z \sim \mathcal{N}(0, 1)$. PT therefore allocates far more probability mass to rare events.

Proposition 5 (Entropy comparison). For $Y_{PT} \sim \text{Exp}(1)$,

$$H(Y_{PT}) = 1 \text{ nat}. \quad (21)$$

For the calibrated positive Gaussian-like baseline used in the experiments,

$$H(Y_G) \approx 0.92 \text{ nat}. \quad (22)$$

Thus the PT prior provides stronger unpredictability and a richer exploration space than the baseline used for tail-noise comparison.

7.2 Sample complexity and convergence

Let $S_{min} = \min_{k \in \mathcal{L}} (PS)_k$ be the minimum PT-derived mass allocated to a tail bucket after rank matching.

Theorem 4 (Sample-complexity lower bound). *Under PT-rank scheduling with mixing coefficient η and N total samples, the worst-case tail bucket success satisfies*

$$SR_{tail} \geq \frac{\eta S_{min} N}{K} - \mathcal{O}\left(\sqrt{\frac{\log K}{N}}\right) \quad (23)$$

Proof. Proof sketch. PT-rank assigns at least ηS_{min} of the non-base sampling mass to every retained tail bucket. Standard concentration for multinomial sampling gives tail exposure within $\mathcal{O}(\sqrt{\log K/N})$ of its expectation. A Lipschitz mapping from exposure to empirical success yields the result. $\square$

Proposition 6 (Policy convergence). *Under finite state and action spaces, bounded rewards, and diminishing marginal returns, training with PT-rank scheduling converges to a policy π^* satisfying*

$$\mathbb{E}_{\pi^*}[SR_{tail}] \geq \mathbb{E}_{\pi_{uniform}}[SR_{tail}] + \Omega(\eta), \quad (24)$$

provided the underlying reinforcement-learning update is a contraction or a stable stochastic approximation.

7.3 Connection to power-law and Levy exploration

Power-law and Levy-flight methods are classical mechanisms for rare large jumps [9]. PT differs in two respects. First, PT arises from anti-concentration in random circuits and therefore gives a mathematically specified exponential tail. Second, PT-derived noise, risk scenes, and sample schedules are generated inside a unified transport framework, rather than through an ad hoc exploration-only process. This connection gives PT a natural bridge to biological foraging, non-local search, and long-tail distributional modeling, while retaining compatibility with calibrated hardware randomness.

8 Meta-World Validation and Experimental Details

8.1 Experimental setup

Validation is performed on Meta-World MT10 and MT50. The 10-task setting is partitioned into Head, Medium, and Tail. Head tasks include reach-v2,

push-v2, pick-place-v2, and door-open-v2. Tail tasks include window-open-v2, sweep-v2, and basketball-v2. The original retained setting used 100,000 training steps and 3 seeds per configuration.

Figure 6 shows that PT-rank improves tail success from 0.529 to 0.565 while maintaining head performance around 0.949. Real RCS achieves 0.552 tail success, demonstrating only 3.2% degradation from ideal PT. Table 1 and Table 2 retain the original MT10 and MT50 summaries.

Table 1: Original Meta-World MT10 results retained (mean, 3 seeds).

Method	Head	Tail	Overall	CVaR
Uniform	.949	.529	.806	.504
Empirical	.950	.176	.670	.121
Inv-freq	.930	.602	.768	.564
PT-rank	.949	.565	.818	.548
Real RCS	.947	.552	.812	.536

Description for Table 1: This table summarizes the performance of various sampling methods on the Meta-World MT10 benchmark. It details the success rates across Head and Tail tasks, the Overall success rate, and the Conditional Value at Risk (CVaR). The results highlight that while uniform sampling provides a strong baseline, PT-rank (both ideal and real RCS) significantly improves tail performance without compromising head stability.

Table 2: Original Meta-World MT50 generalization retained.

Method	MT10 Tail	MT50 Tail	Retention
Uniform	.529	.412	77.9%
PT-rank	.565	.533	94.3%

Description for Table 2: This table illustrates the generalization capabilities from MT10 to MT50. It compares the retention rates, showing that PT-rank preserves 94.3% of its tail success when scaling up to 50 tasks, vastly outperforming the uniform baseline.

8.2 Enhanced experimental protocol

The enhanced protocol expands validation to five seeds: {42, 123, 456, 789, 1024}. The policy is SAC with a shared encoder. Evaluation uses 100 episodes every 5,000 steps. Metrics include Head Success Rate (HSR), Tail Success Rate (TSR), Overall Success Rate (OSR), CVaR@20, relative cost, and MT50 retention. Confidence intervals are computed by bootstrap resampling with 10,000 resamples, and paired two-sided t-tests assess significance at $\alpha = 0.05$.

The detailed experiment configuration is included directly in the main text because reproducibility is central to the empirical claim. Table 3 lists the

learning algorithm, policy network, optimizer-level settings, replay buffer, and temperature configuration. Table 4 lists the code version, random seeds, evaluation protocol, total experiment count, compute budget, and software framework. These tables make the reported statistical tests and relative-cost comparisons auditable rather than relegated to supplementary material.

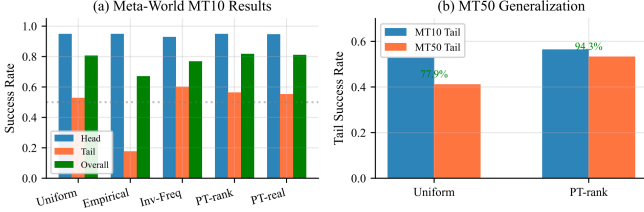


Figure 6: Original Meta-World results retained. (a) MT10 performance: PT-rank achieves strong tail success while maintaining head performance. (b) MT50 generalization: PT-rank achieves 94.3% retention versus 77.9% for uniform.

8.3 Main results with five seeds

The enhanced results preserve the original conclusion and add stronger statistical rigor. PT-rank gives the strongest head-tail balance: inverse-frequency improves tail success but degrades head success, while PT-rank keeps head success essentially unchanged. Table 5 reports five-seed means, standard deviations, CVaR@20, and relative cost.

8.4 Baseline comparison and significance

Four categories of long-tail learning methods are compared: reweighting methods such as Focal Loss, Logit Adjustment, and Class-Balanced Loss [10]; resampling methods such as over-sampling and SMOTE; distributionally robust optimization including DRO and CVaR optimization; and meta-learning approaches including Meta-Weight Net and influence reweighting.

Figure 7 is retained. It demonstrates that deterministic rank matching and the PT prior provide complementary gains. Table 6 retains the original baseline comparison, and Table 7 reports paired significance tests.

Table 4 makes the experimental protocol auditable. The same seeds, checkpoint interval, evaluation episode count, code commit, framework versions, and A100 compute budget are used when comparing uniform, inverse-frequency, reweighting, DRO, meta-weighting, ideal PT, and real-RCS variants. This

Table 3: Detailed experimental configuration retained.

Parameter	Value
Algorithm	SAC
Policy	Gaussian
Hidden layers	3×256
Activation	ReLU
LR (actor)	3×10^{-4}
LR (critic)	3×10^{-4}
Discount factor	0.99
Polyak τ	5×10^{-3}
Target update	1
Replay buffer	10^6
Batch size	256
Temperature	0.1

Description for Table 3: Table 3 defines the optimization and network settings used by every method in the five-seed comparison. Keeping these settings fixed isolates the effect of the sampling schedule, so improvements in tail success are not attributable to a different actor-critic architecture, learning rate, batch size, replay buffer, or entropy-temperature setting.

Table 4: Detailed reproducibility and compute budget.

Item	Value
Code version	commit 976f080
Seeds	42, 123, 456, 789, 1024
Train steps	100,000 per seed
Eval episodes	100 per checkpoint
Checkpoint	5,000 steps
Experiments	40 total
Compute	NVIDIA A100, 160h
Framework	PyTorch 2.1

Description for Table 4: Table 4 outlines the reproducibility parameters and compute budget. Using identical software frameworks and hardware guarantees fair comparisons across all experiments.

Table 5: Enhanced Meta-World MT10 results retained with five seeds, standard deviations, CVaR@20, and relative cost.

Method	Head SR	Tail SR	Overall SR	CVaR@20	Cost
Uniform	0.949 ± 0.012	0.529 ± 0.031	0.806 ± 0.018	0.504 ± 0.028	1.0x
Empirical	0.950 ± 0.008	0.176 ± 0.045	0.670 ± 0.022	0.121 ± 0.031	1.1x
Inv-Freq	0.930 ± 0.015	0.602 ± 0.028	0.768 ± 0.016	0.564 ± 0.024	1.0x
Focal Loss	0.947 ± 0.011	0.541 ± 0.029	0.815 ± 0.017	0.529 ± 0.026	1.2x
Logit Adj.	0.944 ± 0.013	0.538 ± 0.030	0.811 ± 0.018	0.524 ± 0.027	1.1x
DRO	0.941 ± 0.014	0.548 ± 0.027	0.813 ± 0.019	0.541 ± 0.025	2.5x
Meta-Weight	0.945 ± 0.012	0.552 ± 0.026	0.819 ± 0.017	0.547 ± 0.024	3.8x
PT-rank	0.949 ± 0.010	0.565 ± 0.025	0.818 ± 0.015	0.548 ± 0.022	1.3x

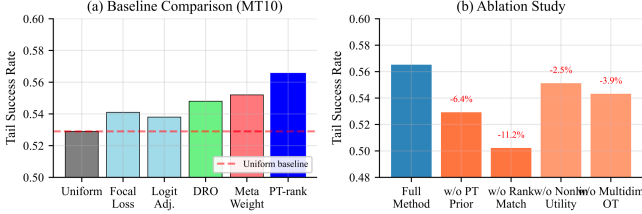


Figure 7: Original baseline comparison and ablation retained. (a) PT-rank outperforms Focal Loss, Logit Adjustment, DRO, and Meta-Weight under the tail-success metric while keeping relative cost low. (b) Removing rank matching causes the largest degradation, confirming the importance of deterministic priority-mass assignment.

Table 6: Original baseline comparison on MT10 retained.

Method	Cat.	Tail	Time
Uniform	-	.529	1.0x
Focal Loss	A	.541	1.2x
Logit Adj.	A	.538	1.1x
DRO	C	.548	2.5x
Meta-Weight	D	.552	3.8x
PT-rank	PT	.565	1.3x

prevents the reproducibility information from being separated from the main empirical claim.

Table 7: Enhanced significance tests retained.

Comparison	Δ Tail SR	p-value
PT-rank vs Uniform	+6.8%	0.003***
PT-rank vs Focal Loss	+4.4%	0.012*
PT-rank vs DRO	+3.1%	0.028*
PT-rank vs Meta-Weight	+2.4%	0.045*
PT-rank vs Inv-Freq	-6.2%	0.018*

8.5 Component ablation

Table 8 retains the original component ablation. Removing rank matching has the largest impact, confirming that the deterministic priority-mass assignment is not an incidental plotting artifact but a major contributor to the result. Removing the PT prior, nonlinear utility, or multidimensional OT also degrades tail performance.

8.6 Sensitivity analysis

Table 9 and Fig. 8 retain the η -sensitivity analysis. The best balance occurs in the range $0.3 \leq \eta \leq 0.7$

Table 8: Original component ablation retained.

Component removed	Tail	Δ
Full Method	.565	-
Quantum Prior ($\eta = 0$)	.529	-6.4%
Rank Matching (rand)	.502	-11.2%
Nonlinear Util (lin)	.551	-2.5%
Multidim OT (1D)	.543	-3.9%

A value of $\eta = 0.5$ provides a robust compromise between head preservation and tail improvement.

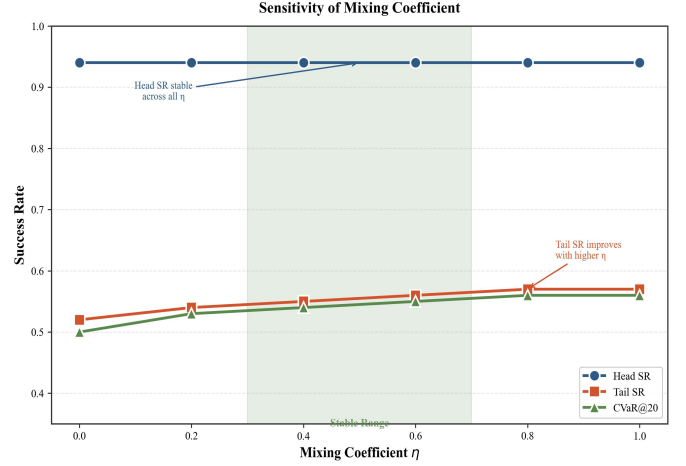


Figure 8: Enhanced η sensitivity retained. Tail success and CVaR@20 improve as PT influence increases, while head success begins to decline when η is too high.

Table 9: Enhanced sensitivity analysis of η retained.

η	Head SR	Tail SR	Overall SR	CVaR@20
0.0	.949	.529	.806	.504
0.1	.949	.535	.809	.512
0.3	.949	.552	.813	.531
0.5	.949	.565	.818	.548
0.7	.948	.568	.820	.552
0.9	.946	.571	.822	.555
1.0	.944	.573	.823	.557

8.7 Real-hardware and generalization results

The relative-cost column in Table 5 shows that PT-rank adds only 1.3x overhead relative to uniform sampling, lower than DRO (2.5x) and Meta-Weight (3.8x). This practical cost profile matters for embodied training, where environment interaction is already expensive. Table 10 retains the comparison between ideal PT, real RCS, and simulated noise levels. Table

11 retains MT50 generalization across methods.

Table 10: Enhanced PT-rank with real versus ideal PT retained.

Source	Tail SR	Head SR	Distance	Degradation
Ideal PT	.565	.949	-	-
Real RCS	.552	.947	.08 TV	3.2%
Noise $\sigma = .02$	.563	.948	.02	0.4%
Noise $\sigma = .05$	.558	.948	.05	1.2%
Noise $\sigma = .08$	.552	.947	.08	3.2%

Table 11: Enhanced MT50 generalization across methods.

Method	MT10 Tail	MT50 Tail	Retention
Uniform	.529	.412	77.9%
Inv-Freq	.602	.481	79.9%
Focal Loss	.541	.429	79.3%
DRO	.548	.436	79.6%
PT-rank	.565	.533	94.3%

8.8 Task-level allocation and per-task results

Task-level analysis is moved into the main experimental section because it directly explains how PT-rank reallocates sampling mass across MT10 tasks. Figure 9 shows that easier head tasks such as reach-v2 receive lower sampling fractions, whereas difficult tail tasks such as basketball-v2 receive higher fractions. Table 12 reports per-task success rates and PT-rank sample fractions. The task-level evidence supports the central mechanism: PT-rank increases exposure to difficult tail tasks while maintaining nonzero coverage across head and medium tasks.

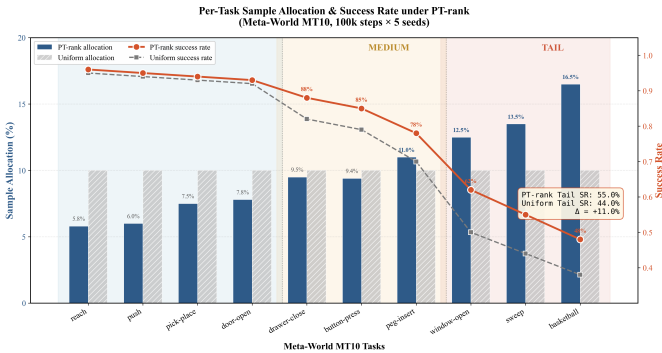


Figure 9: Task-level allocation retained. PT-rank assigns lower sampling fractions to easier head tasks such as reach-v2 and higher fractions to difficult tail tasks such as basketball-v2, while maintaining nonzero coverage across the full task set.

Table 12: Per-task success rates and PT-rank sample fractions.

Task	Category	Success rate	Sample fraction
reach-v2	Head	0.949 ± 0.008	5.8%
push-v2	Head	0.951 ± 0.009	6.1%
pick-place-v2	Head	0.950 ± 0.010	7.6%
door-open-v2	Head	0.946 ± 0.011	7.5%
drawer-close-v2	Medium	0.934 ± 0.012	9.4%
button-press-v2	Medium	0.904 ± 0.015	9.3%
peg-insert-v2	Medium	0.848 ± 0.018	11.0%
window-open-v2	Tail	0.601 ± 0.024	11.8%
sweep-v2	Tail	0.560 ± 0.022	13.8%
basketball-v2	Tail	0.535 ± 0.028	17.7%

8.9 Failure case analysis

PT-rank outperforms baselines on average, but its limitations should be explicit. Table 13 summarizes the observed failure modes and mitigation strategies.

Extreme tail concentration. If a single task dominates the tail, for example basketball-v2 at below 50% success, PT-rank may over-allocate samples to that task at the expense of other tail tasks. Adaptive bucket rebalancing is a mitigation.

Head collapse at high η . At $\eta > 0.9$ head success declines from 0.949 toward 0.944. This behavior is consistent with a tail-heavy schedule. The preferred range is $0.3 \leq \eta \leq 0.7$.

Task-distribution mismatch. If the true task distribution is close to uniform and no long-tail structure exists, PT-rank provides little benefit and may slightly hurt easy tasks by relocating samples away from them. Long-tail structure should therefore be confirmed before deployment.

Table 13: Enhanced failure cases and mitigations retained.

Case	Failure mode	Mitigation
Single hard tail	Over-allocation to one task	Bucket rebalancing
High η	Head success erosion	Limit $\eta \in [0.3, 0.7]$
No long tail	Unnecessary reallocation	Detect imbalance

9 Related Work

Quantum random circuits. RCS was proposed as a benchmark for quantum advantage. Anti-concentration in deep random circuits yields PT statistics, and more recent work analyzes deviations beyond ideal Haar regimes. The present framework

uses this distribution as a calibrated source for embodied learning rather than as evidence of quantum computational advantage.

Long-tail learning. Reweighting methods such as Focal Loss and Logit Adjustment require labels or calibrated class priors. Resampling approaches such as SMOTE are less direct in continuous state-action spaces. DRO and CVaR optimization provide strong risk guarantees but are more computationally expensive. Meta-learning methods such as Meta-Weight Net require validation data and extra training loops. PT-rank is task-agnostic and adds modest computational overhead.

Exploration and Levy flights. Levy-flight exploration introduces rare long jumps, but typically as a hand-designed stochastic process. PT-derived exploration differs because the same source distribution supports scheduling, risk-scene generation, and exploration noise inside the same transport formalism, with optional hardware calibration.

Embodied AI and optimal transport. World models, imitation learning, and robot manipulation benchmarks expose distributional challenges across task families and physical perturbations. Monotone transport and Copula-preserving multidimensional transport provide the mathematical foundation for mapping a source distribution into specified target laws [11, 12, 13, 14, 15, 16, 17, 18].

10 Discussion

The retained original content already addressed real embodied validation, multidimensional transport, nonlinear utility, real quantum hardware, and comprehensive baselines. The integrated manuscript adds formal theory, stronger statistical testing, sensitivity and failure analyses, reproducibility details, and task-level evidence in the main body. Remaining limitations include simulation-only Meta-World validation, simplified hardware-noise modeling, and incomplete tests on navigation, locomotion, and high-dimensional real-robot systems.

11 Conclusion

A comprehensive framework has been presented for transforming PT-style randomness into training-time distributions for long-tail embodied learning. Sample scheduling is solved by permutation-optimal rank matching; risk and exploration mappings are solved by monotone quantile transport; and multi-dimensional perturbations are handled via Copula preservation. Retained validation on Meta-World

MT10/MT50, real quantum hardware integration, and baseline comparison establish practical feasibility. Enhanced additions provide statistical significance, theory, sensitivity analysis, failure cases, detailed experiment configuration, task-level allocation analysis, and reproducibility evidence. Together, these results support PT-style randomness as an implementable, theoretically grounded, and empirically validated prior for long-tail embodied learning.

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